

## COMPSCI 189 HW1

I certify that all solution in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted. — Chenyang Zhang, 3043218909

1. Let  $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 6 \\ 3 & 6 & 9 \end{pmatrix}$ ,  $B = \begin{pmatrix} 1 & 2 & 3 & 6 \\ 2 & 0 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{pmatrix}$

$\therefore \text{rank}(A)=2, \text{rank}(B)=3$

$\therefore$  The problem has no solution.

2.  $\therefore MP=PD$

$\therefore P$  is the eigenvector of  $M$

Let  $V_i$  be the  $i$  column of  $P$

We can get  $MV_i = \lambda_i V_i$

(a):  $MV_i - \lambda V_i = (M - \lambda I)V_i = 0$

Assuming that  $(M - \lambda I)$  is invertible

We can get  $(M - \lambda I)^T (M - \lambda I) \cdot V_i = (M - \lambda I)^T \cdot 0$

$\therefore I \cdot V_i = 0$

$\therefore V_i \neq 0$

$\therefore$  The assumption is invalid

Therefore,  $|M - \lambda I| = 0$

Substituting  $M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ,  $\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$  into  $|M - \lambda I| = 0$

We can get  $(\lambda - 1)(\lambda + 1) = 0$

Therefore,  $\lambda_1 = 1, \lambda_2 = -1$

$MV_1 = \lambda_1 V_1, MV_2 = \lambda_2 V_2$

$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, P^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$

$\therefore M^n = (P D P^{-1})^n = P D^n P^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1^n & 0 \\ 0 & (-1)^n \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$   
 $= \frac{1}{2} \begin{pmatrix} 1+(-1)^n & 1-(-1)^n \\ 1-(-1)^n & 1+(-1)^n \end{pmatrix}$

2.(b): Similarly as (a), let  $\lambda$  be the eigenvalue,  $V$  be the eigenvector

$$MV = \lambda V$$

$$\begin{aligned} |M - \lambda I| &= (2 - \lambda)(-\frac{7}{6} - \lambda) - (-5) \cdot \frac{1}{2} \\ &= \lambda^2 - \frac{5}{6}\lambda - \frac{7}{3} + \frac{5}{2} \\ &= \underline{(2\lambda - 1)(3\lambda - 1)} \end{aligned}$$

$$\therefore \lambda_1 = \frac{1}{2}, \lambda_2 = \frac{1}{3}, D = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$$

$$P \text{ and } P^{-1} \text{ can be } \begin{pmatrix} 10 & 3 \\ 3 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & -3 \\ -3 & 10 \end{pmatrix}$$

$$\therefore M_2 = \begin{pmatrix} 10 & 3 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 & -3 \\ -3 & 10 \end{pmatrix}$$

$$\begin{aligned} M_2^n &= \begin{pmatrix} 10 & 3 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} (\frac{1}{2})^n & 0 \\ 0 & (\frac{1}{3})^n \end{pmatrix} \begin{pmatrix} 1 & -3 \\ -3 & 10 \end{pmatrix} \\ &= \begin{pmatrix} 10 \cdot (\frac{1}{2})^n - (\frac{1}{3})^{n+1} & -3 \cdot (\frac{1}{2})^n + 10 \cdot (\frac{1}{3})^{n+1} \\ 3 \cdot (\frac{1}{2})^n - (\frac{1}{3})^{n+1} & -9 \cdot (\frac{1}{2})^n + 10 \cdot (\frac{1}{3})^{n+1} \end{pmatrix} \end{aligned}$$

2.(c): Similarly as (b):

$$M_3 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

$$\begin{aligned} \therefore M_3^n &= \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3^n & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 2 \cdot 3^n - 2^n & -2 \cdot 3^n + 2^{n+1} \\ 3^n - 2^n & -3^n + 2^{n+1} \end{pmatrix} \end{aligned}$$

$$3. A = U \Sigma V^T$$

$\therefore U$  and  $V$  are orthogonal matrix

$\therefore V^T$  preserves the Euclidean norm of  $X$

$$\therefore \|V^T X\|_2 = \|X\|_2$$

$$\text{Let } Z = V^T X$$

$$\|\Sigma Z\| = \sum_{i=1}^2 \sigma_i^2 z_i^2$$

$$A^T A = \begin{bmatrix} 2 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix}$$

let  $\lambda$  and  $V$  be  $A^T A$ 's eigenvalue and eigenvector ( $\lambda \neq 0, V \neq 0$ )

$$A^T A \cdot V = \lambda V$$

$(A^T A - \lambda I)V = 0$ , assuming that  $A^T A - \lambda I$  is invertible

$$(A^T A - \lambda I)^T (A^T A - \lambda I)V = 0$$

$$\therefore V \neq 0$$

$\therefore$  Assumption fails

$$\text{Therefore, } |A^T A - \lambda I| = 0$$

$$\begin{vmatrix} 8-\lambda & 2 \\ 2 & 5-\lambda \end{vmatrix} = \lambda^2 - 13\lambda + 40 - 4 \\ = (\lambda - 9)(\lambda - 4)$$

$$\therefore \lambda_1 = 9, \lambda_2 = 4$$

$$\sigma_1 = \sqrt{\lambda_1} = 3, \sigma_2 = \sqrt{\lambda_2} = 2$$

$$\Sigma = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$$

$$(A^T A - \lambda_1 I)V_1 = 0 \Rightarrow \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} V_1 = 0$$

$\therefore V$  is orthogonal matrix

$$\therefore V_1 = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$$

$$\text{Similarly: } V_2 = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{pmatrix}$$

$$\|\Sigma Z\|^2 = \sigma_1^2 z_1^2 + \sigma_2^2 z_2^2 \\ = 9z_1^2 + 4z_2^2$$

$$\therefore z_1^2 + z_2^2 = 1$$

$$\therefore z_1 = 1, z_2 = 0, \|\Sigma Z\|_{\max} = 9$$

$$\therefore V = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$$

$$4. X^T X = V \Sigma^T U^T U \Sigma V^T$$

$\therefore V$  and  $U$  are orthogonal matrix

$$\therefore U^T U = I, V^T V = I$$

$$\therefore X^T X = V \Sigma^T \Sigma V^T$$

$$X^T X v_i = V \Sigma^T \Sigma V^T v_i$$

$$= V \Sigma^T \Sigma e_i$$

$$= V \cdot \sigma_i^2 e_i$$

$$= \sigma_i^2 \cdot V e_i$$

$$= \sigma_i^2 v_i$$

Therefore, the columns of  $V$  are the eigenvectors of  $X^T X$

5. (a):  $3 \times 3$

$$(b): T = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$(c): T = \begin{pmatrix} 0 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \end{pmatrix}_{n \times n}$$

$$(d): T = \begin{pmatrix} 0 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \end{pmatrix}$$

$$IT = I_{\text{flip}}$$

6. (a):

$$\frac{\partial f}{\partial x_1} = 3x_1^2 - 3x_2, \quad \frac{\partial f}{\partial x_2} = 3x_2^2 - 3x_1$$

$$\frac{\partial^2 f}{\partial x_1^2} = 6x_1, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = -3, \quad \frac{\partial^2 f}{\partial x_2^2} = 6x_2, \quad \frac{\partial^2 f}{\partial x_2 \partial x_1} = -3$$

6. (b):

$$\frac{\partial f}{\partial x} = 8x - 8y + 4, \quad \frac{\partial f}{\partial y} = 2y - 8x + 6$$

$$\begin{cases} 8x - 8y + 4 = 0 \\ 2y - 8x + 6 = 0 \end{cases} \Rightarrow \begin{cases} x = \frac{7}{6} \\ y = \frac{5}{3} \end{cases}$$

$$\text{When } x = \frac{7}{6}, y = \frac{5}{3}: \frac{\partial^2 f}{\partial x^2} = 8, \quad \frac{\partial^2 f}{\partial y^2} = 2, \quad \frac{\partial^2 f}{\partial x \partial y} = -8$$

$$\therefore \frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} - \left( \frac{\partial^2 f}{\partial x \partial y} \right)^2 = 16 - 64 = -48 < 0$$

$\therefore \left( \frac{7}{6}, \frac{5}{3} \right)$  is not the point that yield a minimum

6.c

$$(i) \frac{\partial f}{\partial x} = e^{xy} \cdot y + 2xy, \quad \frac{\partial f}{\partial y} = e^{xy} \cdot x + x^2$$

$$(ii) \frac{\partial^2 f}{\partial x^2} = e^{xy} \cdot y^2 + 2y, \quad \frac{\partial^2 f}{\partial y^2} = e^{xy} \cdot x^2, \quad \frac{\partial^2 f}{\partial x \partial y} = e^{xy} \cdot xy^2 + e^{xy} + 2x$$

6d.

$$(i): \frac{\partial f}{\partial x} = \frac{2x}{x^2+y^2+1}, \quad \frac{\partial f}{\partial y} = \frac{2y}{x^2+y^2+1}$$

$$(ii): \frac{\partial^2 f}{\partial x^2} = \frac{2(y^2-x^2+1)}{(x^2+y^2+1)^2}, \quad \frac{\partial^2 f}{\partial y^2} = \frac{2(x^2-y^2+1)}{(x^2+y^2+1)^2}, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{-4xy}{(x^2+y^2+1)^2}$$

$$7a: \frac{dz_k}{dz_{k-1}} = W_{n-1}$$

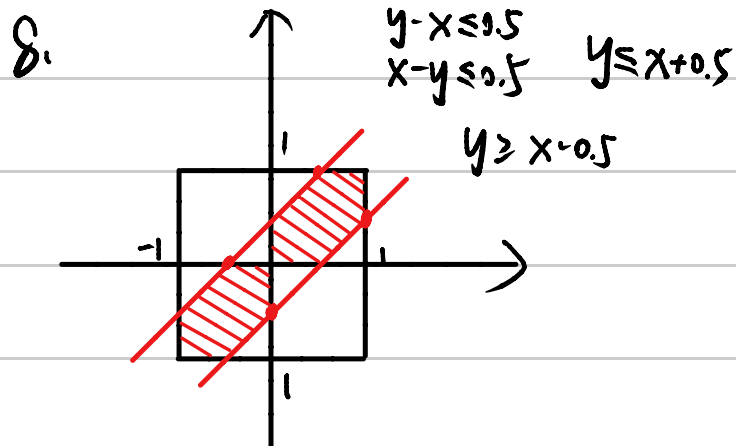
$$7b: \frac{dz_n}{dz_1} = \frac{dz_n}{dz_{n-1}} \cdot \frac{dz_{n-1}}{dz_{n-2}} \cdots \frac{dz_2}{dz_1}$$

$$= W_{n-1} \cdot W_{n-2} \cdots W_1$$

$$= \prod_{i=1}^{n-1} W_i$$

$$7c: \because \frac{dz_2}{db_1} = 1, \quad \frac{dz_n}{dz_2} = \prod_{i=2}^{n-1} W_i$$

$$\therefore \frac{dz_n}{db_1} = \frac{dz_n}{dz_2} \cdot \frac{dz_2}{db_1} = \prod_{i=2}^{n-1} W_i$$



$$\therefore P = \frac{2 \times \frac{3}{4}}{2 \times 1} = \frac{3}{4}$$

$$9(a). P(D > t) = (1-2t)^9$$

9(b).

$$P(D < t) = 1 - (1-2t)^9$$

$$\therefore p(t) = \frac{\partial P(D < t)}{\partial t} = -9(1-2t)^8 \cdot (-2) \\ = 36(1-2t)^8$$

$$E(D) = \int_0^{\frac{1}{2}} t \cdot p(t) dt \\ = \int_0^{\frac{1}{2}} 36t(1-2t)^8 dt \\ = \frac{1}{40}$$

$$\frac{1}{40} \times 360 = 9^\circ$$

Therefore, the expected length is  $9^\circ$

10.

$$P(T=1) = P(T=1|C=1) \cdot P(C=1) + P(T=1|C=0) \cdot P(C=0) \\ = 0.9 \cdot 0.001 + 0.03(1-0.001)$$

$$\therefore P(C=1|T=1) = \frac{P(T=1|C=1) \cdot P(C=1)}{P(T=1)} \\ = \frac{0.9 \cdot 0.001}{0.9 \cdot 0.001 + 0.03 \cdot 0.999} \\ \approx 0.029$$

$$11. P = 420 \times \frac{1}{3} = 140$$